CALCULUS · ANTIDERIVATIVES AND DEFINITE INTEGRALS

Riemann Sum Lab

Explore & Reflect

Name _____ Date _____ Period _____

Open **Riemann Sum Lab** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. For $f(x) = \frac{x^2}{4} + 1$ on $[0, 4]$, will left-endpoint rectangles overestimate or underestimate the area? Why?

2. Can a definite integral be 0 even when the graph encloses area?

2 Explore use the tool

1. $\frac{x^2}{4} + 1$ on $[0, 4]$: record the Left sums for $n = 4$, 20 and 100, and the exact integral.

2. Same function and interval: record the Midpoint sum for $n = 4$. Compare it with the Left sum for $n = 20$.

3. $\sin x$ on $[-3, 3]$: record the exact integral and the total area.

4. $4 - x^2$ on $[2, 4]$: what is the sign of the integral, and why?

3 Record use the tool and fill in the table

Sums and integrals

f(x)	Interval	Method, n	Sum	Exact integral	Over or under?
$\frac{x^2}{4} + 1$	$[0, 4]$	Right, 4			
$4 - x^2$	$[0, 2]$	Left, 4			
$4 - x^2$	$[0, 2]$	Midpoint, 4			
$\frac{x^3}{8} - x$	$[-2, 2]$	Trapezoid, 4			

4 Explain what did you discover?

1. A Riemann sum adds _____ × Δx for each subinterval.
2. As n increases, the sums approach _____.
3. Rectangles below the x-axis count as _____, so the definite integral measures _____ area.
4. For an increasing function, left sums _____ and right sums _____.
5. Explain the difference between $\int_{-3}^{3} \sin x\,dx$ and the total area between sin x and the x-axis on [−3, 3].

5 Apply a new situation, no tool

1. Estimate $\int_{0}^{6} f(x)\,dx$ with a Right sum and n = 3 if f(2) = 4, f(4) = 7 and f(6) = 9.

2. A particle has velocity v(t) = 3 − t m/s for $0 \le t \le 5$. Find its displacement $\int_{0}^{5} v(t)\,dt$ and the total distance it travels.

3. Write $\lim_{n\to\infty} \sum_{i=1}^{n} \left(2 + \frac{3i}{n}\right)^2 \cdot \frac{3}{n}$ as a definite integral.